Floater Finder

Prostate-biopsy fragment dependence

A CONCH safety experiment for pathologist review

Ashby Hackathon 2026

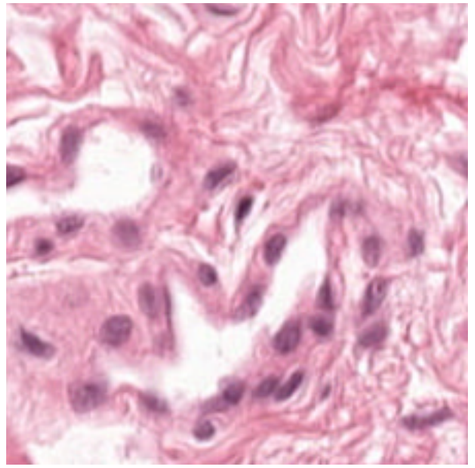
Measured development results. Clinical usefulness remains unverified.

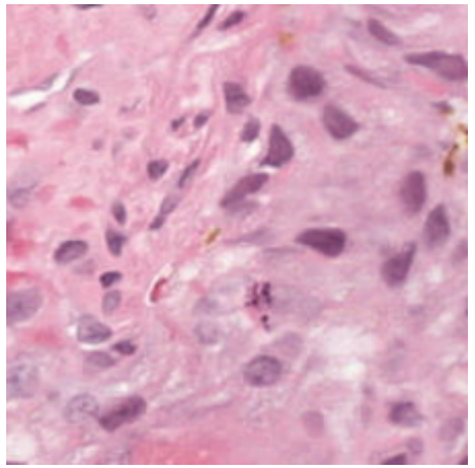
The specimen-consistency problem

Extraneous tissue can create a diagnostic discrepancy

A disease classifier can recognize cancer without establishing that the tissue belongs to the specimen.

The intended action is focused pathologist review. Original tissue and predictions remain intact.





Actual Radboud tiles

Native benign / GP4 annotations

No confirmed floater shown

CONCH and the review question

CONCH supplies reusable visual representations

Fixed morphology prompts support an engineering benign/cancer scorer. They do not certify specimen identity.

Disease prediction

What does the sampled tissue resemble?

Review evidence

Which component is unusual and consequential, and what uncertainty remains?

120 real slides and 50,729 embeddings

40 benign 40 GG1 40 ISUP 4–5

72 training 24 calibration 24 held-out

112 µm tissue fields, spatial coordinates and native mask fractions

512-dimensional normalized CONCH v1 embeddings

119 usable masks. Distinct slide splits, patient identity unknown.

Delta Slurm handled WSI processing and GPU encoding. Subsequent stress tests reuse saved features on CPU.

Clean benign/cancer baseline

Frozen top-5% pooling, calibration only

Original slide label	Predicted benign	Predicted cancer
Benign	5	3
Cancer	1	15

20 / 24

Accuracy 83.3%

Balanced accuracy 78.1%

Only 5 of 8 benign hosts start correct. New contamination errors must use those five as the denominator.

S1: a controlled cancer-fragment stress test

32 benign host slides, 21 reused donor slides

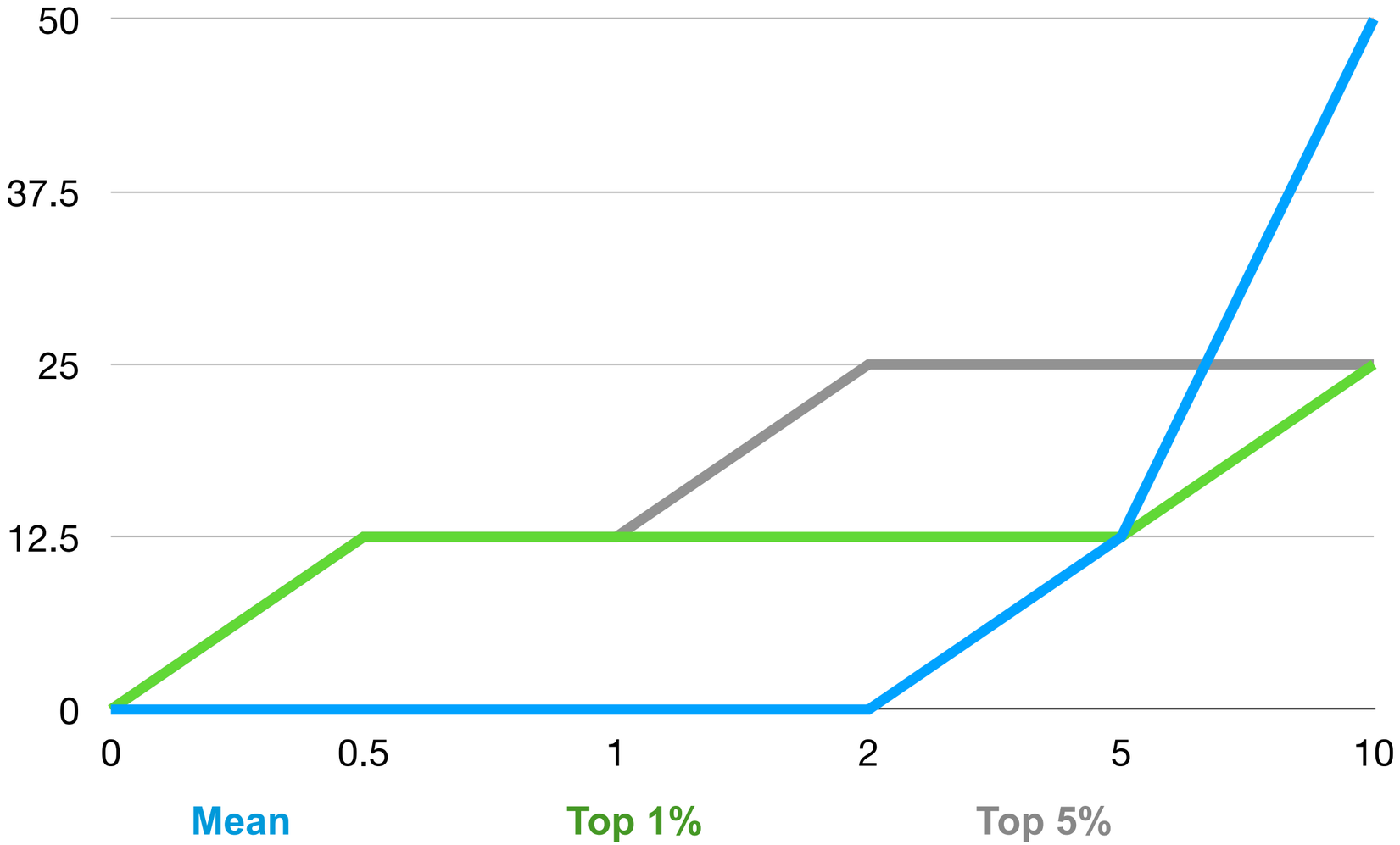
192 trials across 0, 0.5, 1, 2, 5 and 10% doses

Dose = injected tiles / original host tiles, rounded up

Each donor fragment uses unique four-connected saved tiles with ≥10% native cancer annotation per tile.

Synthetic embedding mixtures and translated placement. No assertion of natural floaters or independent patients.

Small fragments can change the prediction



Y: prediction changes (% of 8 calibration hosts). X: requested dose categories (%)

Top-5% new errors

1/5, 1/5, 2/5, 2/5, 2/5

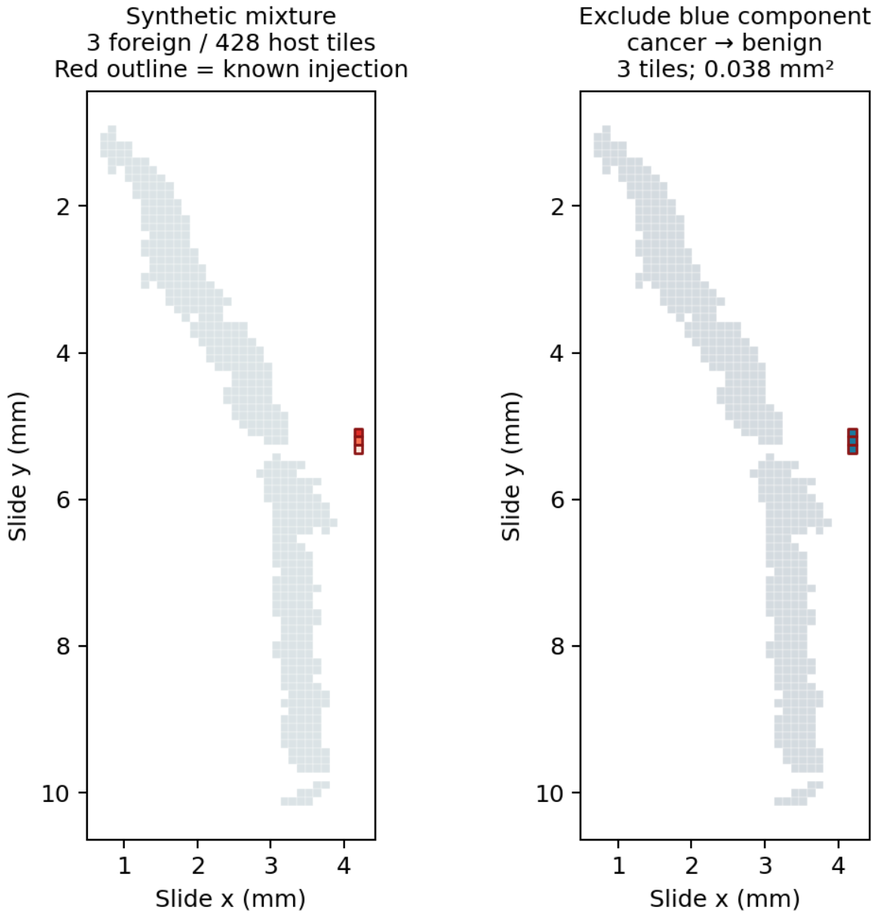
Originally correct benign hosts

Doses 0.5 / 1 / 2 / 5 / 10%

8 error instances across

2 unique calibration hosts

Known synthetic donor component



3 donor tiles

0.0375 mm² injected footprint

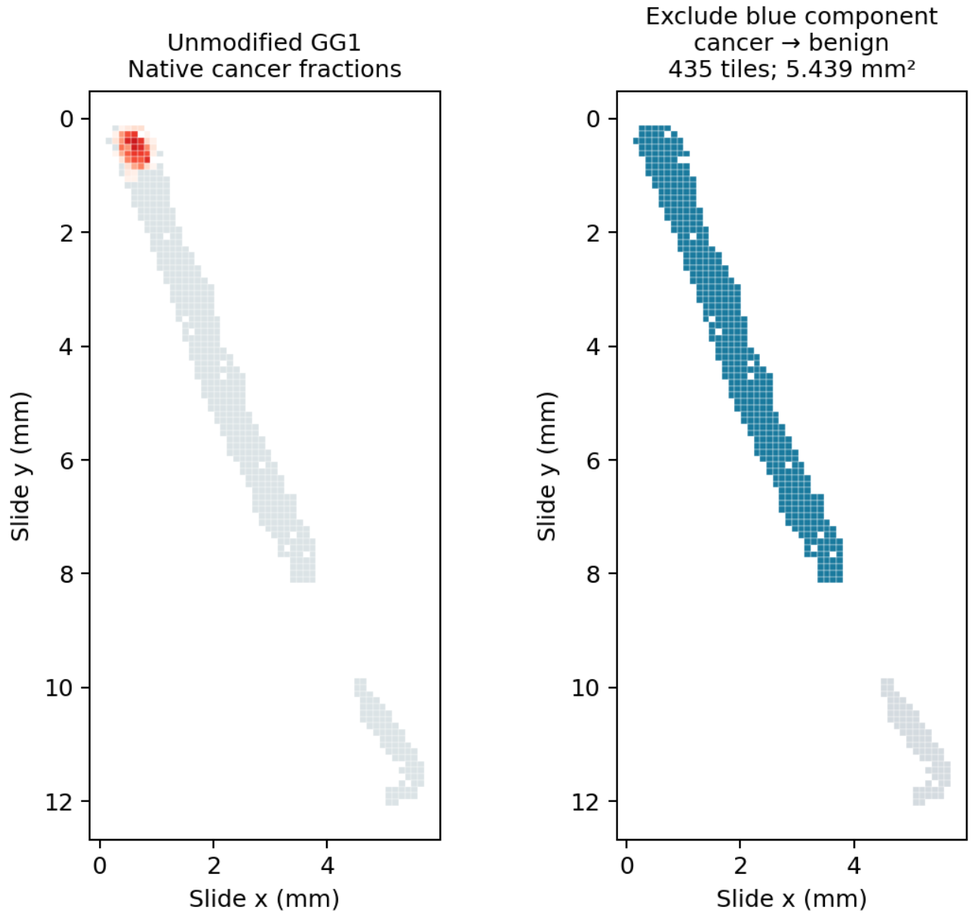
428 original host tiles

Cancer before exclusion

Benign after exclusion

Known injection in a saved-feature mixture. These are tile footprints, not a microscopy composite.

Genuine cancer can also change the decision



Native annotated cancer

0.171 mm² annotation within

a 5.439 mm² sampled footprint

435 tiles in this component

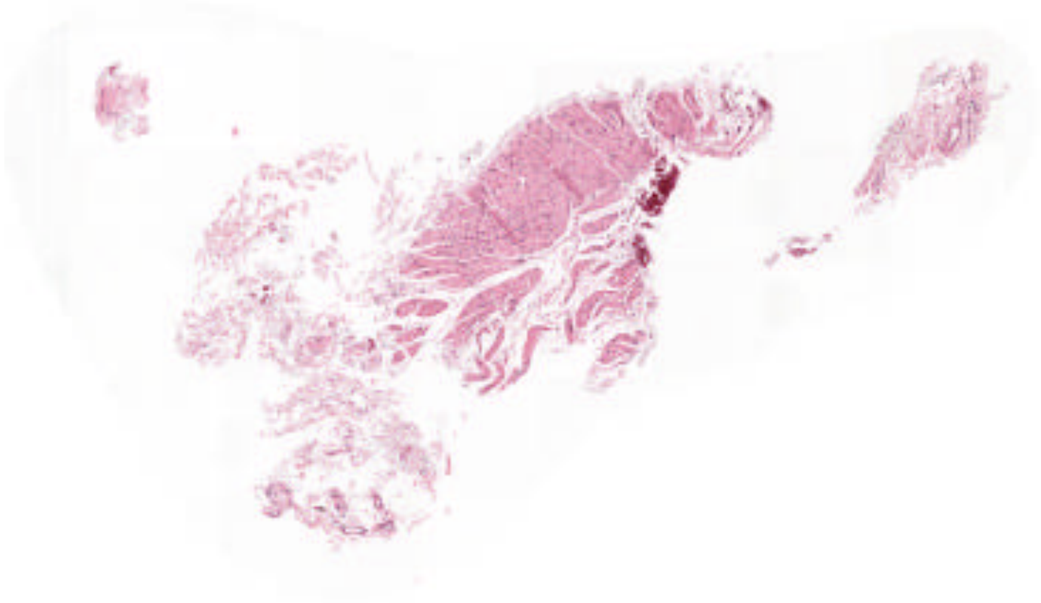
Cancer before exclusion

Benign after exclusion

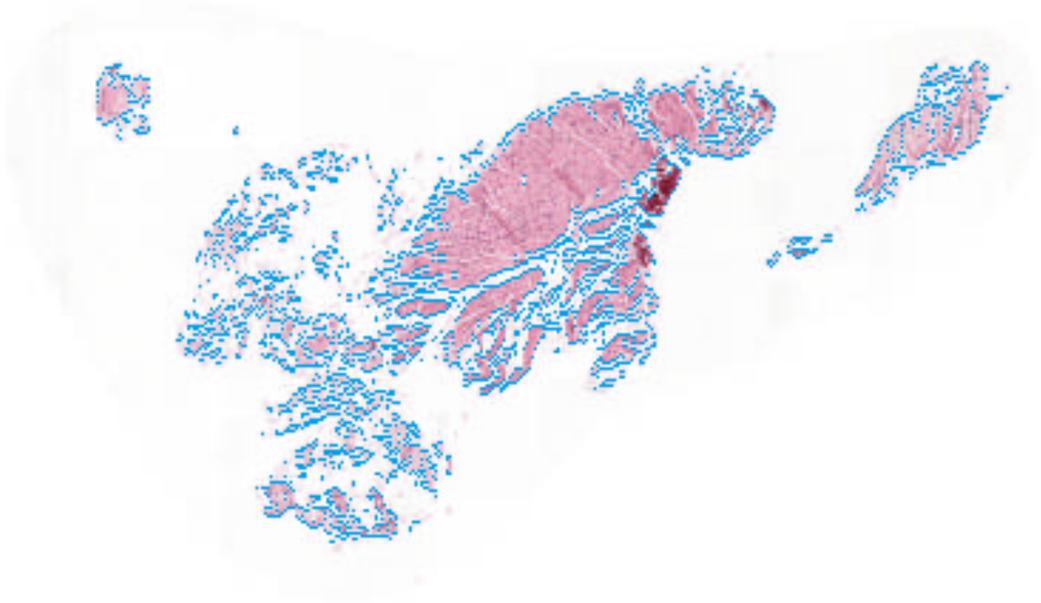
Red shading shows native cancer fraction. Dependence alone does not establish foreignness.

Observed tissue geometry has limits

Actual RGB overview



Observed foreground edge proxy



Worst mapping among training slides: 76% of tiles assigned. Review abstains on this case.

RGB at factor 16, without gap filling. Boundaries are proxies. 4 of 6 formerly single-component calibration cases become multiple.

Review utility conflicts with reviewer burden

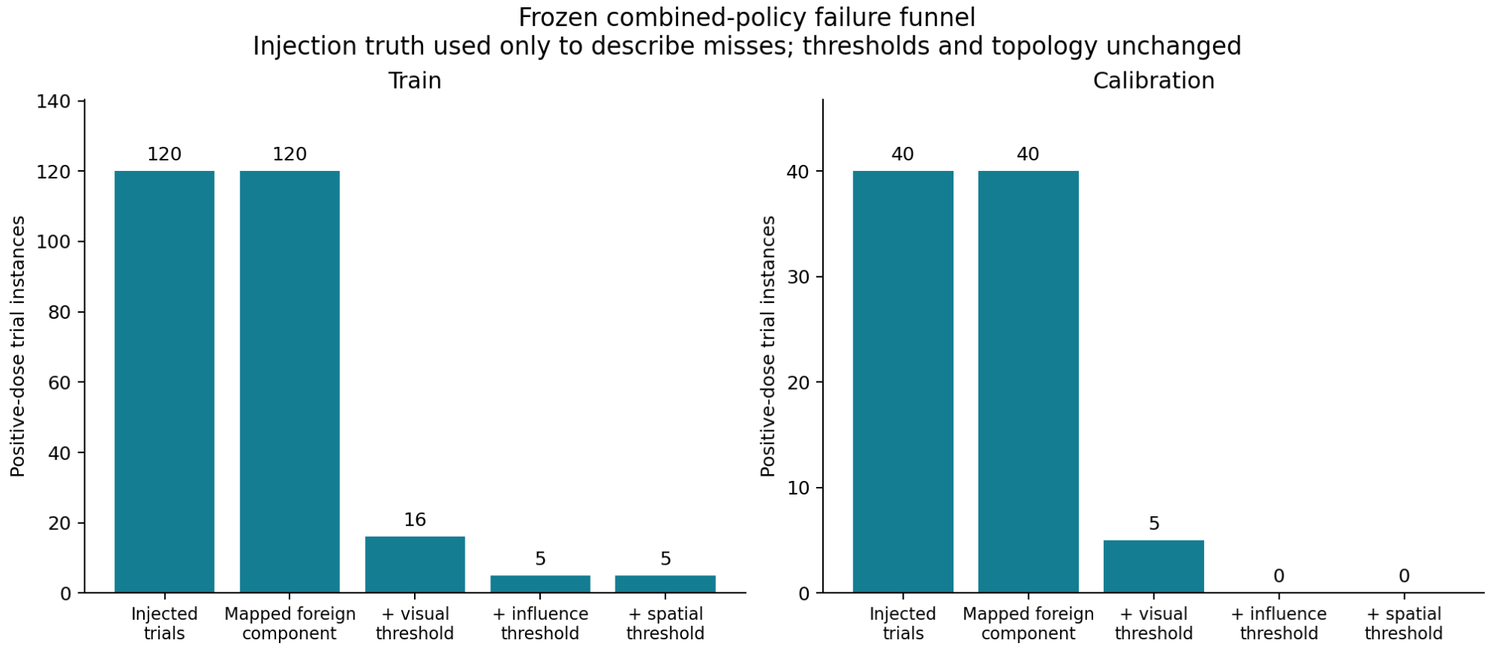
Calibration rule	Fragments hit	Clean slides alerted	Components / slide
Combined	0 / 40	0 / 24	0
Visual only	0 / 40	0 / 24	0
Decision only	8 / 40	4 / 24	0.167
Spatial only	32 / 40	19 / 24	3.583

Development targets: ≤10% clean slides alerted and ≤0.5 components per slide

Decision-only hypothetical filtering damages 3/20 originally correct clean predictions and flags 1/26 native cancer components.

40 mixtures reuse 8 hosts. Clean alerts measure review burden, not adjudicated natural-contaminant specificity.

The combined v1 policy failed its utility gate



0 / 40 fragments detected

No useful candidate among

81 preregistered rules

0/24 alerts has a 95%

Wilson interval of 0–13.8%

The frozen protocol records rejection. Its final-test job remains blocked.

S2: clean grade scorer before contamination

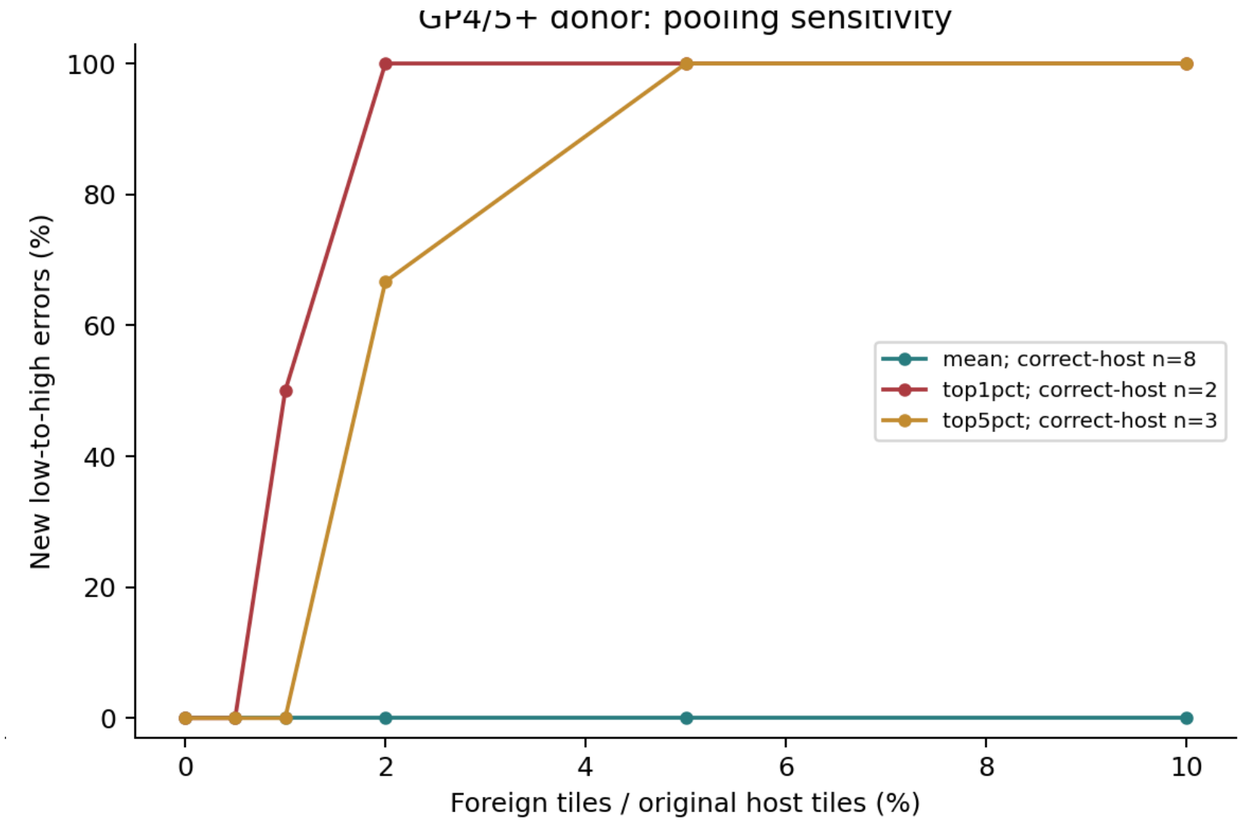
Mean pooling selected before S2 outcomes

Clean low/high scorer	Training correct	Calibration correct
Mean, primary	35 / 48	12 / 16
Top 1%	29 / 48	10 / 16
Top 5%	30 / 48	11 / 16

Primary calibration: low 8/8, high 4/8. Limited high-grade sensitivity remains explicit.

760 trials across four donor conditions. 32 host slides, 37 donor slides. Exact oracle: 760/760.

S2 primary result: no calibration label flips



X: requested foreign / original-host tile percentage (ceil-rounded counts)

Mean: 0 / 8

New low-to-high errors

at every tested GP4/5 dose

Top 1% at 10%: 2/2

Top 5% at 10%: 3/3

Same-grade GP3 control also causes 1/3 top-5% new errors at 10%. A binary flip is not a clinical Grade Group upgrade.

Detector v2: a separately versioned ranking study

Training selects rank threshold and review cap

Visual anomaly and prediction influence become separate rank features. Geometry contributes observed area and isolation.

Native annotation, injection truth and donor identity enter evaluation only. They never enter the blinded review features.

Known calibration outcomes make this iterative development, not independent validation.

Ranked review still fails useful calibration recall

Ranked-review condition	Fragments hit	Clean slides alerted	Clean damage
Selected combined, train	22 / 120	3 / 72	3 / 61
Selected combined, known cal	0 / 40	1 / 24	0 / 20
Top-one visual + relevance, cal	21 / 40	24 / 24	2 / 20

Unbudgeted top-one review flags 14 native cancer components. It exceeds both review-burden targets.

No supported v2 policy and no validated useful improvement over v1. The 24-slide final test remains blocked.

What the experiment demonstrates

Demonstrated

Verified PANDA-to-CONCH pipeline

S1 synthetic model-sensitivity effects

S2 null primary calibration outcome

Two review policies rejected honestly

Still unverified

Naturally occurring floater detection

Patient-level independence

Clinical cancer or grading validity

Independent final-test performance

Preserving a negative result is part of the safety check.

Next evidence requires a new scientific gate

24 final-test slides remain sealed

S3: useful review with genuine-cancer preservation and realistic controls.

S4: external institutions and models, after separate approval and embedding-compatibility checks.

A useful new policy needs a new frozen protocol and explicit final-test authorization.
Additional models or GPU work require a separate scope.

Research prototype. Pathologist review only. No automatic tissue removal.

Sources and reproducibility

Clinical context

Sapp et al. 2021. doi:10.14309/crj.0000000000000583

Foundation model

Lu et al. Nature Medicine 2024. doi:10.1038/s41591-024-02856-4

Dataset

Bulten et al. Nature Medicine 2022. doi:10.1038/s41591-021-01620-2

Code and measured development evidence

github.com/MuneebHoda/floater-finder/tree/feature/panda-tiling

S1/v1 source: 06c7b77. S2 source: 35e0a8. v2 source: 668d9b8.

Media hashes, source reports and exact protocol paths accompany this deck.